Deep Learning (CAI-466)

Objectives:

- **RAG**
- **RAG and LLM limitations**
- **RAG key stages**
- **RAG vs Semantic Search**
- **Benefits of RAG**

Retrieval-augmented generation (RAG) :

Retrieval-augmented generation (RAG) is a technique that enables generative artificial intelligence (Gen AI) models to retrieve and incorporate new information. It modifies interactions with a large language model (LLM) so that the model responds to user queries with reference to a specified set of documents, using this information to supplement information from its pre-existing training data. This allows LLMs to use domain-specific and/or updated information. Use cases include providing chatbot access to internal company data or generating responses based on authoritative sources.

RAG improves large language models (LLMs) by incorporating information retrieval before generating responses. Unlike traditional LLMs that rely on static training data, RAG pulls relevant text from databases, uploaded documents, or web sources. According to Ars Technica, *"RAG is a way of improving LLM performance, in essence by blending the LLM process with a web search or other document look-up process to help LLMs stick to the facts."* This method helps reduce AI hallucinations, which have led to real-world issues like chatbots inventing policies or lawyers citing nonexistent legal cases.

By dynamically retrieving information, RAG enables AI to provide more accurate responses without frequent retraining. According to IBM, *"RAG also reduces the need for users to continuously train the model on new data and update its parameters as circumstances evolve. In this way, RAG can lower the computational and financial costs of running LLM-powered chatbots in an enterprise setting."*

Beyond efficiency gains, RAG also allows LLMs to include source references in their responses, enabling users to verify information by reviewing cited documents or original sources. This can provide greater transparency, as users can cross-check retrieved content to ensure accuracy and relevance. The term "retrieval-augmented generation" (RAG) was first introduced in 2020 by Douwe Kiela et al.

RAG and LLM Limitations

In June 2024, Ars Technica reported, *"LLMs aren't humans, of course. Their training data can age quickly, particularly in more time-sensitive queries. In addition, the LLM often can't distinguish specific sources of its knowledge, as all its training data is blended together into a kind of soup."* In 2023, during its launch demonstration, Google's Bard provided incorrect information about the James Webb Space Telescope, an error that contributed to a $100 billion decline in Alphabet's stock value.

RAG systems do not inherently verify the credibility of the sources they retrieve from, which can lead to misleading or inaccurate responses. AI systems can generate misinformation even when pulling from factually correct sources if they misinterpret the context. MIT Technology Review gives the example of an AI-generated response stating***, "The United States has had one Muslim president, Barack Hussein Obama."*** The model retrieved this from an academic book titled Barack Hussein Obama: America's First Muslim President? and misinterpreted the content, generating a false claim.

Retrieval-Augmented Generation (RAG) is a method that allows large language models (LLMs) to retrieve and incorporate additional information before generating responses. Unlike LLMs that rely solely on pre-existing training data, RAG integrates newly available data at query time. Ars Technica states, *"The beauty of RAG is that when new information becomes available, rather than having to retrain the model, all that's needed is to augment the model's external knowledge base with the updated information."*

The BBC describes ***"prompt stuffing"*** as a technique within RAG, in which relevant context is inserted into a prompt to guide the model's response. This approach provides the LLM with key information early in the prompt, encouraging it to prioritize the supplied data over pre-existing training knowledge.

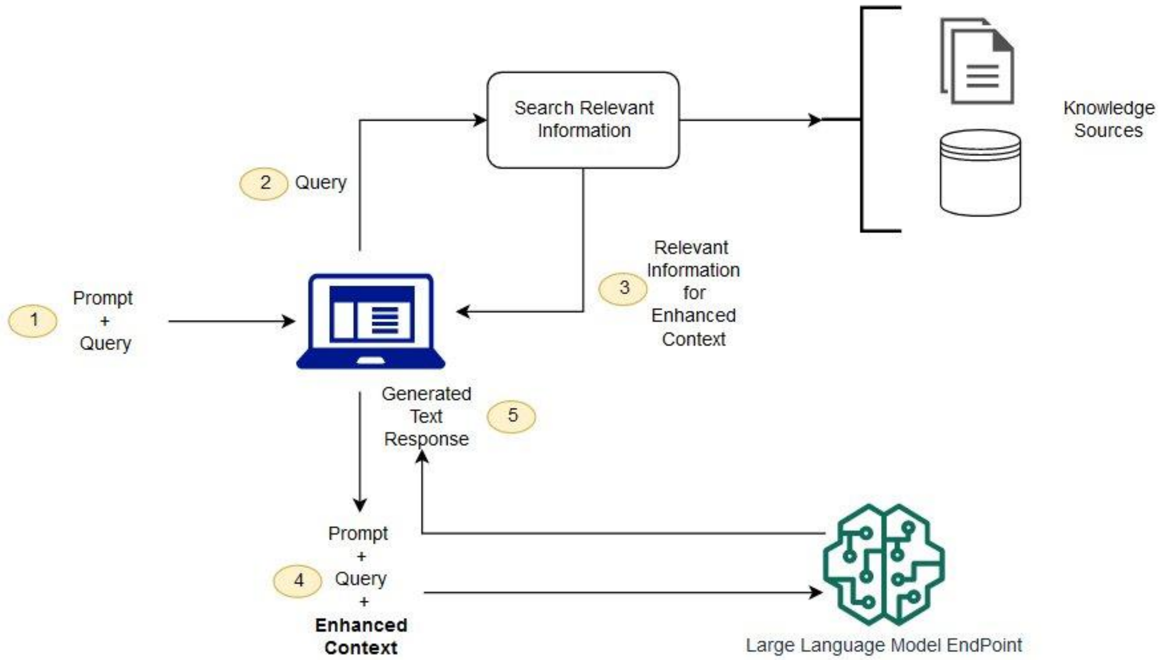
Process

Retrieval-Augmented Generation (RAG) enhances large language models (LLMs) by incorporating an information-retrieval mechanism that allows models to access and utilize additional data beyond their original training set. AWS states, *"RAG allows LLMs to retrieve relevant information from external data sources to generate more accurate and contextually relevant responses"* **(indexing)**. This approach reduces reliance on static datasets, which can quickly become outdated. When a user submits a query, *RAG uses a document retriever to search for relevant content from available sources before incorporating the retrieved information into the model's response* **(retrieval)**. Ars Technica notes that *"when new information becomes available, rather than having to retrain the model, all that's needed is to augment the model's external knowledge base with the updated information"* **(augmentation)**. *By dynamically integrating relevant data, RAG enables LLMs to generate more informed and contextually grounded responses* **(generation)**. IBM states that "in the generative phase, the

LLM draws from the augmented prompt and its internal representation of its training data to synthesize an engaging answer tailored to the user in that instant.

RAG key stages

The following diagram shows the conceptual flow of using RAG with LLMs.



Indexing

Typically, the data to be referenced is converted into LLM embeddings, numerical representations in the form of a large vector space. RAG can be used on unstructured (usually text), semi-structured, or structured data (for example knowledge graphs). These embeddings are then stored in a vector database to allow for document retrieval.

Retrieval

Given a user query, a document retriever is first called to select the most relevant documents that will be used to augment the query. This comparison can be done using a variety of methods, which depend in part on the type of indexing used.

Augmentation

The model feeds this relevant retrieved information into the LLM via prompt engineering of the user's original query. Newer implementations (as of 2023) can also incorporate specific augmentation modules with abilities such as expanding queries into multiple domains and using memory and self-improvement to learn from previous retrievals.

Generation

Finally, the LLM can generate output based on both the query and the retrieved documents. Some models incorporate extra steps to improve output, such as the re-ranking of retrieved information, context selection, and fine-tuning.

Retrieval-Augmented Generation vs. Semantic Search

RAG isn't the only technique used to improve the accuracy of LLM-based generative AI. Another technique is semantic search, which helps the AI system narrow down the meaning of a query by seeking deep understanding of the specific words and phrases in the prompt.

Traditional search is focused on keywords. For example, a basic query asking about the tree species native to France might search the AI system's database using "trees" and "France" as keywords and find data that contains both keywords—but the system might not truly comprehend the meaning of trees in France and therefore may retrieve too much information, too little, or even the wrong information. That keyword-based search might also miss information because the keyword search is too literal: The trees native to Normandy might be missed, even though they're in France, because that keyword was missing.

Semantic search goes beyond keyword search by determining the meaning of questions and source documents and using that meaning to retrieve more accurate results. Semantic search is an integral part of RAG.

Benefits of Retrieval-Augmented Generation

RAG techniques can be used to improve the quality of a generative AI system's responses to prompts, beyond what an LLM alone can deliver. Benefits include the following:

- The RAG has access to information that may be fresher than the data used to train the LLM.
- Data in the RAG's knowledge repository can be continually updated without incurring significant costs.
- The RAG's knowledge repository can contain data that's more contextual than the data in a generalized LLM.
- The source of the information in the RAG's vector database can be identified. And because the data sources are known, incorrect information in the RAG can be corrected or deleted.

Challenges of Retrieval-Augmented Generation

Because RAG is a relatively new technology, first proposed in 2020, AI developers are still learning how to best implement its information retrieval mechanisms in generative AI. Some key challenges are

- Improving organizational knowledge and understanding of RAG because it's so new
- Increasing costs; while generative AI with RAG will be more expensive to implement than an LLM on its own, this route is less costly than frequently retraining the LLM itself

- Determining how to best model the structured and unstructured data within the knowledge library and vector database
- Developing requirements for a process to incrementally feed data into the RAG system
- Putting processes in place to handle reports of inaccuracies and to correct or delete those information sources in the RAG system